

# WILL DUENKEL

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## Vanderbilt University

**Class of 2028**

3.67 GPA | Bachelor of Engineering & Master of Science, Electrical and Computer Engineering

## Internship Experience

### **Electrical Engineering Intern – Basis**

**July 2026 – Present**

- Building a TransU-Net based deep learning model to disaggregate household power draw into appliance-level load consumption, achieving 92% preliminary accuracy to prototype a novel real-time energy monitoring system for future integration into Basis' smart circuit breakers.

### **Radiation Effects Intern – Institute for Space & Defense Electronics**

**May – July 2026**

- Built a SiC MOSFET double-pulse test and DC-DC buck converter platform to study TID (total-ionizing dose) & SEE (single-event effect) radiation-induced degradation in SiC power devices.
- Diagnosed and resolved hardware faults during platform bring-up.
- Executed double-pulse testing on SiC MOSFETs and captured switching current/voltage transient waveforms. Identified parasitic inductance-induced ringing as dominant error source in switching-loss measurements and proposed corrective hardware roadmap to guide future development.

## Extracurricular Experience

### **New Zealand Undergrad Space Challenge (NZUSC) – Electrical & Technical Lead**

**Aug 2026 – Present**

- Photonics/Electrical Lead on an interdisciplinary team in NZ's national CubeSat challenge; winning payload selected from each university is sent to orbit aboard shared 3U CubeSat via Rocket Lab.
- Designing the photonic and electronic architecture for a NASA Artemis-inspired optical communications payload to transmit HD photo/video for LEO disaster monitoring.

### **Rocket Team (VADL) – Hardware, Avionics, and Outreach Teams**

**Aug 2025 – Present**

- Competing in the NASA Student Launch Competition (team 7× first place, 3× second, 1× third).
- Conducted an avionics trade study that cut costs by ~\$1,300, replacing an expensive IMU with a \$10 altimeter that met all electrical specifications while increasing altitude precision by 12x.
- Resolved critical Parachute Detachment System faults. Mitigated a critical latch hardware fault & implemented moving average filter to smooth altimeter noise, ensuring 100% deployment reliability.
- Co-authoring detailed engineering reports (PDR/CDR) for NASA Student Launch, documenting design tradeoffs, risk analyses, and requirement verification for review by NASA engineers.
- Collaborating with 30+ cross-functional team members.

### **SCALE – Undergraduate Student Member**

**Oct 2025 – Present**

- Selected to participate in the SCALE microelectronics workforce development network.

## Personal Projects

### **SiC MOSFET Imaging**

**Nov – Dec 2025**

- Characterized the elemental structure of an irradiated SiC MOSFET used in aerospace applications for the Institute for Space & Defense Electronics, utilizing FIB-SEM and EDS techniques.

### **AI Powered IMSA Race Car Image Classifier**

**Jan – Nov 2025**

- *Best Project Presentation – 2025 Vanderbilt IEEE Project Showcase.*
- Uses ChatGPT-4o Vision and EasyOCR to classify racecars; 2000+ images with 89% accuracy.

## Skills

**Hardware & Instrumentation:** Oscilloscopes, Rogowski coils, power supplies, signal generators, semiconductor device parameter analyzers, SEM & TEM.

**Software & EDA:** LTSpice, Python, Java, Bash, Xilinx Vivado, KiCad, LaTeX.